

Kunal Avghade

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Summary

Full Stack Software Engineer with **3+ years** building cloud-native distributed systems. Specialist in **Java/Spring Boot, Python/FastAPI, and TypeScript/NestJS**, delivering fault-tolerant platforms and **petabyte-scale** migrations. AWS Certified Solutions Architect with strong algorithmic foundations.

Technical Skills

Languages: Java, Python, TypeScript, JavaScript, C++, SQL

Frameworks: Spring Boot, NestJS, FastAPI, React.js, Next.js

Distributed Systems: Temporal, Kafka, RabbitMQ, gRPC, REST

Cloud/DevOps: AWS, GCP, Docker, Kubernetes, Jenkins, ArgoCD

Databases: PostgreSQL, Redis, MongoDB, Milvus

Professional Experience

Software Development Engineer - II | Calfus Inc.

Jul 2023 – Present

Full Stack Development & Cloud-Native Engineering

Pune, India

• Aetherion — Agentic AI Platform

Tech Stack: Python, TypeScript, Java, FastAPI, Temporal, PostgreSQL, AWS, Docker, LLMs

- Built production-grade **Python, TypeScript, and Java SDKs** enabling developers to author, deploy, and manage AI agents with standardized abstractions for memory, tools, and execution graphs.
- Designed a conversational agent-generator that dynamically translates natural language requirements into executable workflows and prompt configurations using **LLMs**.
- Architected distributed orchestration using **Temporal**, guaranteeing fault-tolerant execution of long-running agent workflows with automated state persistence and recovery mechanisms.
- Optimized asynchronous task pipelines and tool invocation layers, lowering execution latency for concurrent, multi-tenant agent runs.

• NetApp Data Migrator — ONTAP Storage Migration Platform

Tech Stack: Java, Spring Boot, NestJS, Temporal, gRPC, RabbitMQ, Redis, PostgreSQL, Keycloak, GCP, AWS, Azure, ONTAP

- Designed a scalable distributed migration platform for **NetApp ONTAP** capable of handling migrations up to **1+ PB** and **1+ Billion** files across **NFS/SMB** environments. Successfully migrated **5+ TB** and **70+ Million** files of unstructured data within **30 hours** with zero data loss during performance validation.
- Architected resumable, fault-tolerant migration workflows using **Temporal** across distributed worker clusters, supporting automated checkpointing and dead-letter recovery.
- Implemented highly parallel discovery engines and low-latency task coordination using **gRPC, RabbitMQ, and Redis** for dynamic workload partitioning.
- Integrated **Keycloak** RBAC and built real-time operational telemetry to monitor cluster throughput, file-level checksum validation, and active worker health.

• Asurint — Background Verification Microservices

Tech Stack: Java, Spring Boot, GCP, Kafka, ArgoCD, Jenkins, PostgreSQL

- Scaled **GCP**-based microservices automating background check workflows, processing millions of indexing and verification requests.
- Decoupled heavy PDF report generation from data ingestion pipelines using **Apache Kafka** event-driven messaging.

Key Project

Vision-Based Intelligent Recipe Recommender

Tech Stack: Python, Django, YOLOv8, OpenCV, React, PostgreSQL, JWT

Jan 2023 – May 2023

- Trained a custom **YOLOv8** model to detect food ingredients from raw images, achieving **92% mAP**.
- Built a secure **Django REST API** and **React.js** client with **JWT**, mapping detected items to 5,000+ recipes via optimized SQL joins.

Achievements & Certifications

- **LeetCode:** Ranked in the **Top 7%** globally (Max Rating: 1900+); Solved **1000+** DSA problems. | Profile
- **Drive Dynamo Award:** Won **3 consecutive years** (2023, 2024, 2025) for exceptional technical delivery and leading platform adoption at Calfus Inc.
- **AWS Certified Solutions Architect – Associate & AWS Cloud Practitioner.**
- **IBM AI Engineering Professional & Stanford Machine Learning Specialization.**

Education

Vishwakarma Institute of Information Technology

B.Tech in Electronics & Telecommunication (Software Systems Focus)

Pune, India

2020 – 2023